E-In Son

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Research Interests

My research focuses on learning-based navigation for robots in real-world environments, spanning traversability estimation in unstructured terrain, visual navigation, and world models. Going forward, I aim to develop world models and navigation policies that generalize across environments and robot embodiments.

Education

Seoul National University **Mar 2021 – Aug 2027 (Expected)**
Ph.D. Student, Dept. of Electrical & Computer Engineering *Seoul, Korea*
Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

Konkuk University **Mar 2015 – Feb 2021**
B.S., Electrical & Electronics Engineering *Seoul, Korea*
Summa Cum Laude (GPA 4.33 / 4.5).

Publications

[1] **Feeling Terrain Before Crossing: World Models for Off-Road Navigation.**
E-In Son, Dong-Wook Kim, Ji-Hoon Hwang, Kangsun Lee, Jisung Bae, Jung-Taak Kim, Seung-Woo Seo.
Under review, 2026. [Paper] [Code]

[2] **Which Terrain Is Better? Preference Learning with VLM Prototypes for Off-Road Traversability Ranking.**
Ji-Hoon Hwang, Jisung Bae, E-In Son, Dong-Wook Kim, Jung-Taak Kim, Seung-Woo Seo.
Under review, 2026. [Paper]

[3] **CanonNav: Disentangling Navigation Behavior from Camera Geometry in Cross-Platform Visual Navigation.**
Dong-Wook Kim, Ji-Hoon Hwang, E-In Son, Mintaek Oh, Seung-Woo Seo.
Under review, 2026. [Paper]

[4] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**
E-In Son*, Jung-Taak Kim*, Seung-Woo Seo.
European Conference on Computer Vision (ECCV), 2026. [Paper] [Project]

[5] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**
Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2025. [Paper] [Code]

[6] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**
Yurim Jeon, E-In Son, Seung-Woo Seo.
IEEE International Conference on Robotics and Automation (ICRA), 2024. [Paper] [Code]

[7] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**
Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2024. [Paper]

[8] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**
Se-Wook Yoo*, E-In Son*, Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2024. [Paper] [Project]

Awards & Honors

Earth Rovers Challenge (1st Place) Oct 2024
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
Real-world autonomous navigation competition, AI policies vs. human teleoperators.

Academic Excellence Scholarship 2018 – 2020
Konkuk University

Research Projects

Challengeable Future Defense Technology R&D Program (Phase 2) 2024 – 2026
Agency for Defense Development (ADD)
Leading the planning stack, from learning-based path prediction to real-time safe replanning, for an end-to-end visual navigation system operating in unknown, hazardous terrain.

Challengeable Future Defense Technology R&D Program (Phase 1) 2022 – 2023
Agency for Defense Development (ADD)
Developed the path tracking module for accurate and stable trajectory following.

Teaching Experience

Seoul National University, Teaching Assistant

SNU in Silicon Valley Program, joint with UC Berkeley *(Summer 2024)*
Fundamentals of Random Variables and Probability Processes *(Spring 2023)*
Network Protocol and Design *(Fall 2021, 2023)*

Academic Service

Conference Reviewer: ICRA, IROS, ECCV
Journal Reviewer: RA-L

Skills

Tools and Languages	Python, C/C++, PyTorch, ROS
Communication	Korean (native), English

Reference

Advisor: Prof. Seung-Woo Seo sseo@snu.ac.kr Website